Thm. Cauchy Integral Formula.

Suppose f is Holomorphic on some open set containing a closure of disk D.

If C is a curve with positive orientation of ∂D, then for any $z_0 \in D$,

$$f(z_0) = \frac{1}{2\pi i} \cdot \int_C \frac{f(z)}{z - z_0}\, dz$$



Pf. Let $z_0 \in D$ be fixed, and consider a curve $\Gamma_{\delta,\varepsilon}$ as

Since $\dfrac{f(z)}{z - z_0}$ is Holomorphic on $D \setminus \{z_0\}$, therefore

$$\int_{\Gamma_{\delta,\varepsilon}} \frac{f(z)}{z - z_0}\, dz = 0.$$ Taking $\delta \to 0$, then we obtain

$$\lim_{\delta \to 0} \int_{\Gamma_{\delta,\varepsilon}} \frac{f(z)}{z - z_0}\, dz = \int_{C_\varepsilon} \frac{f(z)}{z - z_0}\, dz + \int_C \frac{f(z)}{z - z_0}\, dz = 0.$$

Meanwhile, $\dfrac{f(z)}{z - z_0} = \dfrac{f(z) - f(z_0)}{z - z_0} + \dfrac{f(z_0)}{z - z_0}$ and since f is Holomorphic,

the first term is bounded, therefore $\lim\limits_{\varepsilon \to 0} \int_{C_\varepsilon} \dfrac{f(z) - f(z_0)}{z - z_0}\, dz = 0.$

Now, $\lim\limits_{\varepsilon \to 0} \int_{C_\varepsilon} \dfrac{f(z_0)}{z - z_0}\, dz = f(z_0) \cdot \lim\limits_{\varepsilon \to 0} \int_{C_\varepsilon} \dfrac{1}{z - z_0}\, dz$

$$= f(z_0) \cdot \lim_{\varepsilon \to 0} \left(-\int_0^{2\pi} \frac{1}{\varepsilon \cdot e^{it}} \cdot \varepsilon \cdot i \cdot e^{it}\, dt \right)$$

$$= -f(z_0) \cdot 2\pi i.$$

Consequently, $\displaystyle\int_C \frac{f(z)}{z - z_0}\, dz = 2\pi i\, f(z_0).$

Corollary. $\displaystyle\int_C \frac{f(z)}{(z - z_0)^{n+1}}\, dz = 2\pi i \cdot \frac{f^{(n)}(z_0)}{n!}$

Proof. Using induction for n. For $n = 0$, clear. If $f^{(n-1)}(z_0) = \dfrac{(n-1)!}{2\pi i} \cdot \displaystyle\int_C \frac{f(z)}{(z - z_0)^n}\, dz$, then

$$\frac{f^{(n-1)}(z_0 + h) - f^{(n-1)}(z_0)}{h} = \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{h} \left[\frac{1}{(z - z_0 - h)^n} - \frac{1}{(z - z_0)^n} \right] dz$$

$$= \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{h} \cdot \left[\frac{h}{(z - z_0 - h)(z - z_0)} \cdot \left(\sum_{k=0}^{n-1} \frac{1}{(z - z_0 - h)^{n-1-k}} \cdot \frac{1}{(z - z_0)^k} \right) \right] dz$$

$$\to \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{(z - z_0)^2} \cdot \sum_{k=0}^{n-1} \frac{1}{(z - z_0)^{n-1}}\, dz \quad \text{as } h \to 0$$

$$= \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot n \cdot \frac{1}{(z - z_0)^{n+1}}\, dz$$

$$= \frac{n!}{2\pi i} \cdot \int_C \frac{f(z)}{(z - z_0)^{n+1}}\, dz = f^{(n)}(z_0).$$

Thm. Cauchy inequality.

$|f^{(n)}(z_0)| \le \dfrac{n!}{R} \cdot \sup\limits_C |f|$ if f is Holomorphic in an open containing $\overline{B_R(z_0)}$, $C = \partial B_R(z_0)$.

Thm. Suppose f is Holomorphic on an open Ω containing a closed disk D having center z_0,

$$f(z)=\sum_{n=0}^{\infty}\frac{f^{(n)}(z_0)}{n!}\cdot(z-z_0)^n \quad (z\in D)$$

Pf. $f(z)=\frac{1}{2\pi i}\cdot\int_C\frac{f(\mu)}{\mu-z}\,d\mu$ $\left(\frac{1}{\mu-z}=\frac{1}{\mu-z_0-(z-z_0)}=\frac{1}{\mu-z_0}\cdot\frac{1}{1-\left(\frac{z-z_0}{\mu-z_0}\right)}\right)$

$$=\frac{1}{2\pi i}\cdot\int_C f(\mu)\cdot\sum_{n=0}^{\infty}\frac{1}{(\mu-z_0)^{n+1}}\cdot(z-z_0)^n\,d\mu$$

$$=\sum_{n=0}^{\infty}\left(\frac{1}{2\pi i}\cdot\int_C\frac{f(\mu)}{(\mu-z_0)^{n+1}}\,d\mu\right)\cdot(z-z_0)^n$$

$$=\sum_{n=0}^{\infty}\frac{f^{(n)}(z_0)}{n!}\cdot(z-z_0)^n.$$

Corollary, Liouville's Theorem.
If $f:\mathbb{C}\to\mathbb{C}$ is entire and bounded, then it is constant.
Proof. Given $z_0\in\mathbb{C}$ and $R>0$, $|f'(z_0)|\le\frac{1}{R}\cdot\sup_{C_R}|f|\to 0$ as $R\to\infty$. Therefore, $f'=0$.
Thus f is constant.

Fundamental Theorem of Algebra.
Non-constant polynomial $P(z)=a_nz^n+a_{n-1}z^{n-1}+\cdots+a_0\in\mathbb{C}[z]$ has a root in $\mathbb{C}$.
Proof. Suppose that $P(z)$ has no root in $\mathbb{C}$. Then, $1/P(z)$ is well-defined on $\mathbb{C}$, entire.
observe that $\frac{P(z)}{z^n}=a_n+\left(\frac{a_{n-1}}{z}+\cdots+\frac{a_0}{z^n}\right)$ $(z\neq 0)$.

The terms $a_{n-k}\cdot z^{-k}$ $(k=1,\cdots,n)$ converges to zero as $|z|\to\infty$, thus for some $R>0$,

$$|P(z)|\ge 2|a_n|\cdot z^n \quad (|z|>R)$$
$$\ge 2|a_n|\cdot R^n$$

Therefore it is bounded below in $|z|>R$, and clearly in $|z|\le R$.
Now, $\frac{1}{P}$ is bounded, thus constant. Contradiction.